

Phase 1 — Core Fundamentals ★★★★★ (Highest Priority)

These are the topics you'll use every day in backend development, AI, and coding interviews.

Topic	Priority	Recommended Problems
Variables & Data Types	★★★★★	20
Operators	★★★★★	25
Strings	★★★★★	70
Lists	★★★★★	80
Tuples	★★★★	25
Sets	★★★★	25
Dictionaries	★★★★★	60
Booleans	★★★★	20
Type Casting	★★★	15
None	★★★	10

Total: ~350 problems

Phase 2 — Logic Building ★★★★★

Conditional Statements

Practice

- Positive/Negative
- Odd/Even
- Largest of 2

- Largest of 3
- Smallest of 3
- Leap Year
- Grade Calculator
- BMI Category
- Age Category
- Login Validation
- ATM PIN
- Electricity Bill
- Tax Calculator
- Discount Calculator
- Triangle Type
- Quadratic Equation

Target: 40 problems

Loops

Practice

- Factorial
- Fibonacci
- Prime Numbers
- Armstrong Numbers
- Perfect Numbers
- Palindrome Numbers
- Reverse Numbers
- Sum of Digits
- Product of Digits
- Multiplication Tables
- Number Patterns
- Star Patterns
- Diamond Patterns
- Pascal Triangle
- Floyd Triangle

Target: 60 problems

Phase 3 — Functions

Practice

- Calculator
- Temperature Converter
- Area Calculator
- Recursive Factorial
- Recursive Fibonacci
- GCD
- LCM
- Default Arguments
- Keyword Arguments
- Variable-length Arguments (`*args`, `**kwargs`)
- Lambda Functions
- Higher-order Functions
- Decorators
- Closures
- Generators

Target: 50 problems

Phase 4 — Strings

Practice

- Reverse String
- Palindrome
- Count Vowels
- Count Consonants
- Character Frequency
- Remove Duplicates
- Remove Spaces
- Capitalize Words
- Word Count
- String Compression
- Find Longest Word
- Replace Words
- Email Validator
- Password Validator
- Caesar Cipher

Target: 70 problems

Phase 5 — Lists ★★★★★

Practice

- Maximum
- Minimum
- Second Largest
- Second Smallest
- Reverse
- Rotate
- Merge
- Remove Duplicates
- Frequency Count
- Bubble Sort
- Selection Sort
- Insertion Sort
- Linear Search
- Binary Search
- Matrix Addition
- Matrix Multiplication

Target: 80 problems

Phase 6 — Tuples ★★★★★

Practice

- Packing
- Unpacking
- Swapping Variables
- Nested Tuples
- Tuple Methods
- Counting Elements
- Index Finding

Target: 20 problems

Phase 7 — Sets

Practice

- Union
- Intersection
- Difference
- Symmetric Difference
- `issubset()`
- `issuperset()`
- `isdisjoint()`
- Duplicate Removal
- Set Comprehension

Target: 25 problems

Phase 8 — Dictionaries

Practice

- Student Database
- Employee Database
- Phone Book
- Inventory
- Nested Dictionaries
- Word Counter
- Character Counter
- Merge Dictionaries
- Sort by Keys
- Sort by Values
- Rename Keys
- Invert Dictionary

Target: 60 problems

Phase 9 — Comprehensions

Practice

- List Comprehension
- Nested List Comprehension
- Dictionary Comprehension
- Set Comprehension
- Conditional Comprehension

Target: 25 problems

Phase 10 — Lambda / Functional Programming ★★★★★

Practice

- `lambda`
- `map()`
- `filter()`
- `reduce()`
- `sorted(key=...)`
- `zip()`
- `enumerate()`
- `any()`
- `all()`

Target: 25 problems

Phase 11 — Iterators & Generators ★★★★★

Practice

- Custom Iterator
- Generator Functions
- Generator Expressions

- `yield`
- `next()`
- `iter()`

Target: 20 problems

Phase 12 — Modules

Practice

Use:

- `math`
- `random`
- `datetime`
- `os`
- `sys`
- `collections`
- `itertools`
- `functools`

Target: 20 problems

Phase 13 — Date & Time

Practice

- Age Calculator
- Countdown
- Difference Between Dates
- Calendar Generator
- Stopwatch

Target: 15 problems

Phase 14 — Math

Practice

- GCD
- LCM
- Square Root
- Power
- Trigonometry
- Combinations
- Permutations

Target: 15 problems

Phase 15 — JSON

Practice

- Python → JSON
- JSON → Python
- Nested JSON
- Read JSON File
- Write JSON File
- Pretty Print JSON

Target: 20 problems

Phase 16 — Regex

Practice

Extract

- Email
- Phone
- URL
- Date
- Price

Validate

- Email
- Password
- IPv4
- Bangladesh Mobile Number
- Username

Replace

- Extra Spaces
- HTML Tags
- Multiple Spaces
- Numbers

Split

- CSV
- Sentences

Target: 35 problems

Phase 17 — Exception Handling



Practice

- Division by Zero
- ValueError
- TypeError
- IndexError
- KeyError
- FileNotFoundError
- Multiple Exceptions
- Custom Exceptions
- `else`
- `finally`

Target: 25 problems

Phase 18 — String Formatting

Practice

- f-Strings
- `.format()`
- Alignment
- Padding
- Number Formatting
- Currency Formatting
- Date Formatting

Target: 15 problems

Phase 19 — Mini Projects

Build at least these:

1. Calculator
 2. Student Management System
 3. Library Management System
 4. Contact Book
 5. Expense Tracker
 6. Quiz App
 7. ATM System
 8. Banking System
 9. Inventory Management
 10. Employee Management
 11. Password Generator
 12. Number Guessing Game
 13. Rock Paper Scissors
 14. To-Do List
 15. Hotel Booking (Console)
 16. Movie Ticket Booking
 17. Shopping Cart
 18. Attendance System
 19. Weather App (using an API later)
 20. JSON-based Database
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Phase 20 — Coding Challenges



Practice on:

- HackerRank (Python)
- LeetCode (Easy → Medium)
- Codewars
- Exercism
- Beecrowd

Target:

- 150 Easy problems
 - 100 Medium problems
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